Angled launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use components to analyze an angled projectile.

SUCCESS CRITERIA - I can...

1. Separate the initial velocity into two components.
2. Use vertical motion to determine flight time.
3. Explain what must be true for your range comparisons to work.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Downward velocity changes from **3** to **8 m/s** in **0.5 s**. **Find acceleration** and **name one possible measurement uncertainty**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $t_{top} = v_{y,i}/g$; $t_{flight} = 2\ v_{y,i}/g$; range = $v_x\ t_{flight}$. Upward initial component $v_{y,i}$; g = **10 m/s²**. Equal launch/landing height and no air resistance; CPA components are supplied.

Worked reasoning: A projectile has initial components **4 m/s** horizontal and **10 m/s** upward. With g=**10 m/s²** it reaches the top in **1 s** and returns to launch height in **2 s**; range=**8 m**.

Your turn: Sketch its path and velocity arrows at launch, top and return.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

At the highest point, **which velocity** component is zero and which remains? Is acceleration zero?



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

A ball launches with velocity components **6 m/s** horizontal and **10 m/s** upward. It lands at its launch height.
1. Use g = **10 m/s²** and ignore air resistance.
2. **Label the diagram**, then **find time** to the top, total flight time and range.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

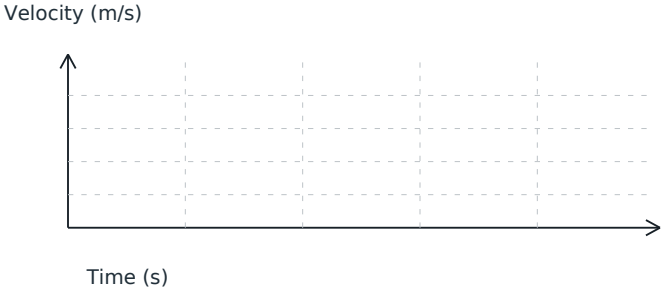
05 Explain your reasoning

In this ideal projectile calculation, why can you double the time to the top only when the object lands at its launch height?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **7 kg** cart starts from rest under **63 N** net force right for **6 s**. **Find acceleration** and final velocity. **State one thing you** are treating as true.
Label the given and predicted quantities on the diagram.



Marzano check - circle: 1 2 3

07 Exit check

Given v_x=**4 m/s**, initial v_y=**15 m/s** up, find time to top and total flight time to launch height. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**. Keep the upward component **10 m/s** but double the horizontal component. **Compare flight time**, maximum height and range. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3